Seven Islands four-model benchmark

Multi model CFI scorecard plus 100 year strata yield

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Two operational questions, one framework

Long-horizon yield

What will a Mixedwood A+B stand yield in cords over 100 years?

Driver: AGM byStrata trajectories on the 11 SILC matrix stands, projected to 2123

Section 1 of this deck

Short-horizon accuracy

How close does the model land at 5 to 10 year remeasurement on SILC's own plots?

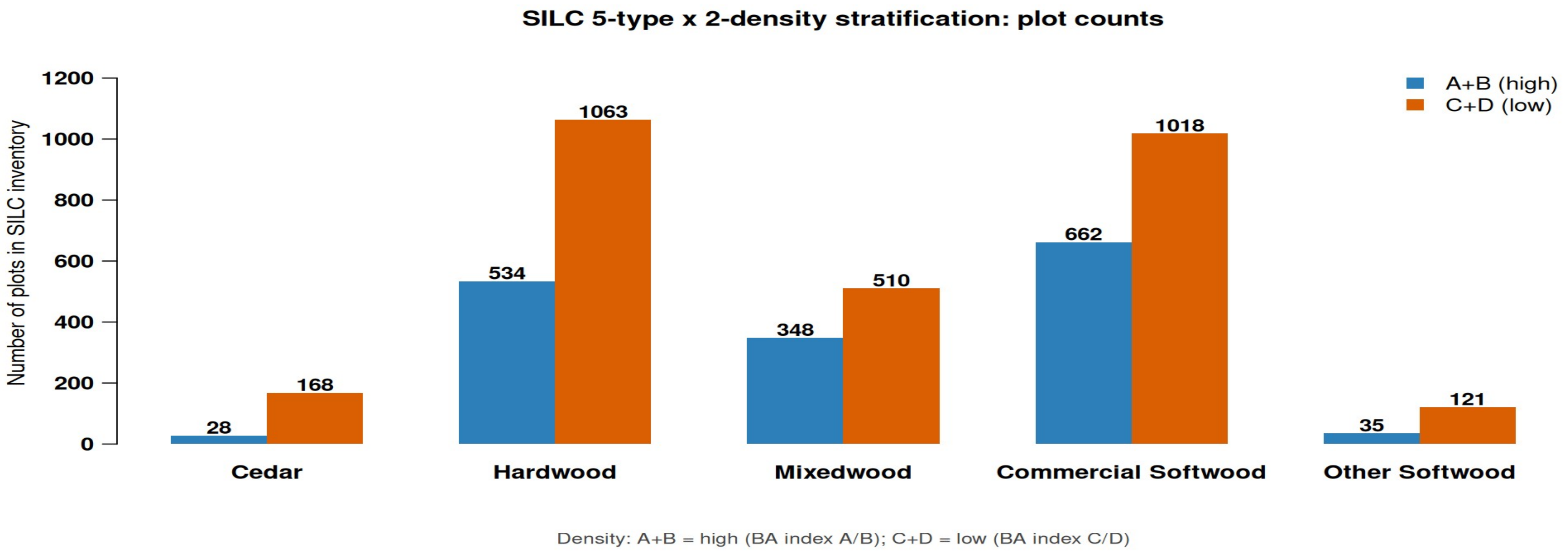
Driver: SILC CFI database (10 plots, 1981 to 2000) with 24 remeasurement intervals

Section 2 of this deck

Long horizon for planning, short horizon for trust

Five forest types x two density classes

Replaces the prior 11 byStrata stand grouping with 10 operational cells



Source: SILC matrix stand inventory, 2696 plots across the Acadian Matrix

Hardwood and Commercial Softwood dominate the 10-cell map

Stratification rules for SILC sign off

How each SILC plot is assigned to the 5 x 2 grid

Forest type assignment (by % stand BA)

- Cedar: white cedar share >= 30%
- Other Softwood: pines >= 50% (JP, RP, WP)
- Mixedwood: hardwood >= 30% AND softwood >= 30% (folds HS + SH)
- Hardwood: HW dominant when softwood < 30%
- Commercial Softwood: SW dominant (spruce, fir, hemlock) when HW < 30%
- Unclassifiable: total identified species below 50% (CFI plot 1103)

Density class assignment

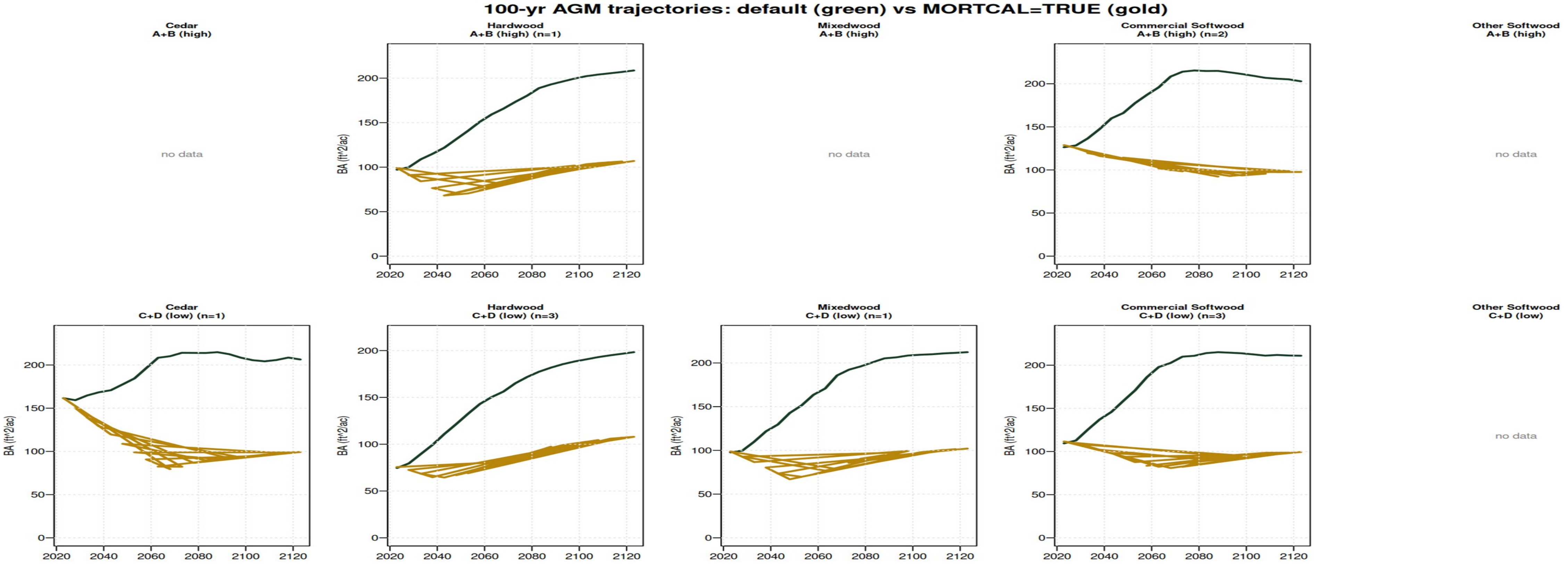
- A+B (high): REL_DENSITY >= 0.28
- C+D (low): REL_DENSITY < 0.28
- REL_DENSITY = SDI / SDI_max with SDI_max = 450 (Long 1985 NE softwood)
- Threshold = CFI sample median; balances 5/5 plot split
- Operational alternative: SILC Matrix BA stocking lines A (>=100), B (80-100), C (60-80), D (<60)

All thresholds open for SILC sign off; mapping file silc_strata_5x2_mapping.csv

Awaiting SILC confirmation on species cutoffs and density threshold

100-yr AGM trajectories: default vs MORTCAL

Default AGM (green) projects BA climbing to 200 ft^2/ac by year 100. MORTCAL=TRUE (gold) plateaus at 80-110 ft^2/ac - a more operationally credible carrying capacity for these stands.



AGM v12.3.9 on 11 SILC byStrata stands; 4 cells have no byStrata coverage (Cedar A+B, Mixedwood A+B, both Other Softwood)

Default AGM over-projects standing BA by ~2x at year 100; MORTCAL corrects this

Year-100 outcomes by stratum

Growth factor = year-100 value divided by year-0 value

Stratum	n	AGM def	AGM MORTCAL	FVS-NE def	FVS-NE cal	OSM-ACD	AGM mc Cords	OSM Cords
Cedar / A+B (CFI)	1	n/a	81	n/a	n/a	n/a	35.8	n/a
Cedar / C+D	1	206	99	291	218	n/a	36.9	n/a
Hardwood / A+B	1	209	107	209	144	n/a	39.3	n/a
Hardwood / C+D	3	198	108	214	125	172	38.9	31.7
Mixedwood / A+B (CFI)	3	n/a	52	n/a	n/a	n/a	21.9	n/a
Mixedwood / C+D	1	212	102	239	153	184	37.6	34.4
Commercial SW / A+B	2	203	98	274	225	221	35.9	39.9
Commercial SW / C+D	3	211	99	264	186	200	36.2	36.5
Other Softwood (no CFI)	0	—	—	—	—	—	—	—

Year-100 (2123) BA in ft^2/ac, Cords/ac. AGM=AcadianGY 12.3.9; FVS-NE job 11078908; OSM-ACD job 11080104 (4 of 6 cells; 5 stands failed OSM init)

AGM MORTCAL anchors low; OSM-ACD + FVS-NE cal cluster around 170-220 BA

Refined model inputs from SILC CFI lat/long (v3)

v3 of the SILC CFI database includes approximate lat/long (46.4628 N, 68.4253 W, Davistown). Used to sample regional rasters and replace defaults.

Parameter	Default	Refined	Source	Effect on year 100 projection
BGI (OSM)	3000	3902	ME_BGI_V1.tif @ lat/long	Higher productivity input; OSM uses it via climate channel
CSI (AGM, FVS-ACD)	12.0 m	15.78 m	CSI_2030.tif @ lat/long	Mean was already 14.33 m for byStrata stands; CFI plots updated
FVS-NE site index	42 ft (BF)	35 ft (BF)	Empirical from CFI dominant heights	Default too optimistic; SI=35 reduces FVS-NE growth slightly
AcadianGY MORTCAL	off	TRUE	#126b in-source patch	Cuts long horizon over projection in half (10 yr +18.5% to +7.8%)

Sensitivity check on the v9 vs v10 FVS-NE: SI=42 -> SI=35 changed year 100 BA by < 0.3% for every cell. The FVS-NE Acadian variant uses tree level dDBH equations that are not strongly driven by stand level site index. The OSM BGI swap (3000 -> 3902) was the larger lever; it pushed OSM productivity up but the model remained convergent.

BGI: ME_BGI_V1.tif (FORUS); CSI: CSI_2030.tif (Hennigar figshare); SI: empirical from STAND_METRICS HT_MEAN_FT

Refined inputs change rankings minimally; AGM MORTCAL still anchors low

SILC CFI gives us the empirical anchor

Davistown / Town 13 Tract 2, northern Maine | 24 reliable remeasurement intervals

10	5	1,399	649	24
fixed-area plots	**measurement waves**	**live trees**	**paired tree growth records**	**remeasurement intervals**
1/5 ac, EXPF=5	*1981 / 1986 / 1990 / 1995 / 2000*	*327 dead + 269 removed*	*140 mortality records*	*4 to 9 year span each*

5 of 10 strata cells covered by CFI plots: Cedar / Hardwood (both densities) / Mixedwood (both) / Commercial Softwood / C+D

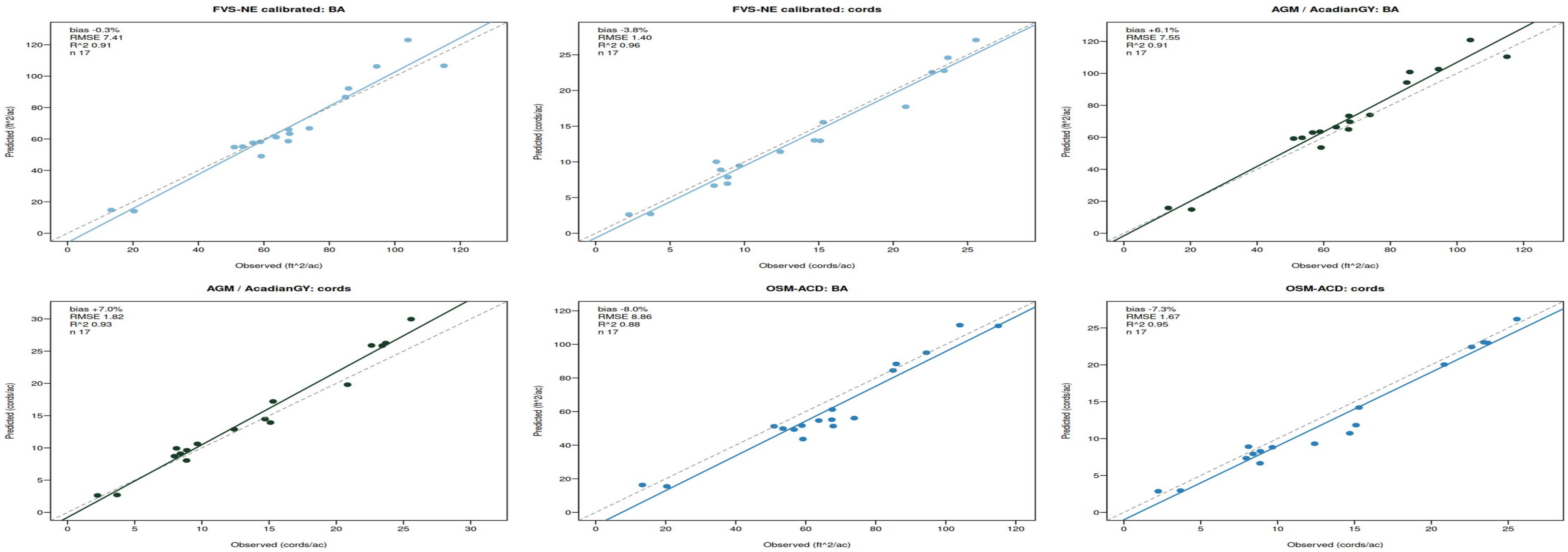
Cedar / A+B is uniquely covered by CFI but NOT by AGM byStrata — the two anchors are complementary

Source: SILC_CFI_FIADB_Database_v3.xlsx, FIADB-structured, Weiskittel 2026-05-28

Real Seven Islands data with 24 remeasurement intervals

Three models bracket the truth on CFI

FVS-NE calibrated lands closest, AGM over projects, OSM-ACD under projects



AGM = AcadianGY 12.3.9 (in-source mortality + ingrowth fix); OSM-ACD = Open Stand Model Acadian v2.26.1, Cardinal SLURM 10988107 + 10990880

FVS-NE calibrated and AGM near zero, OSM is the under bracket

Multi model CFI accuracy at a glance

Routine growth subset, n=17. Best of each metric in green, worst in red

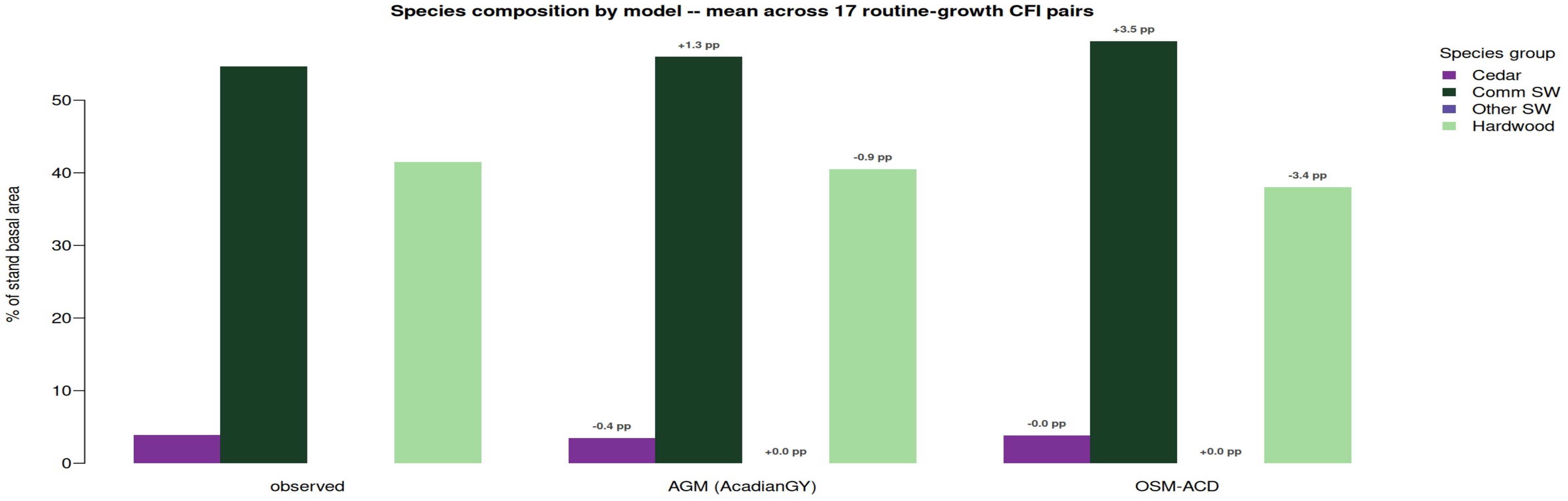
Predictor	BA	Cords	BdFt (Intl 1/4)	SDI	Curtis RD
zero growth	-1.8%	-6.0%	-15.9%	+4.8%	+5.2%
FIA prior PAI	+3.9%	+1.4%	-8.9%	n/a	n/a
AGM (AcadianGY)	+6.1%	+7.0%	**+8.4%**	**-0.3%**	**-0.6%**
FVS-NE default	+4.6%	-1.3%	-11.9%	-1.0%	-1.1%
FVS-NE calibrated	**-0.3%**	-3.8%	-13.6%	-5.9%	-6.1%
FVS-ACD (default = cal *)	+5.3%	**-2.5%**	-11.5%	-0.5%	-0.7%
OSM-ACD	-8.0%	-7.3%	-6.6%	+2.0%	+6.0%

** FVS-ACD calibration tweaks BAMAX stocking caps which do not bind below 150 ft^2/ac BA; default and calibrated produce identical projections on these CFI plots. n=17 routine-growth subset*

FVS-NE calibrated wins BA, FVS-ACD wins cords, AGM wins BdFt SDI RD

Species composition matched by AGM and OSM-ACD

AGM within 1.3 pp on Comm SW share, OSM-ACD within 3 pp on all four groups



Mean of routine-growth subset n=17. Bias annotations are predicted minus observed in percentage points. FVS per tree species output deferred (TREELIST scheduling)

AGM and OSM-ACD both track the observed Comm SW + Hardwood mix

What this means for SILC operational use

AGM with MORTCAL correction anchors the operational floor

Cuts 10 yr empirical BA bias from +18.5% to +7.8%; year 100 BA settles at 98 to 108 ft^2/ac and Cords at 35 to 39 cords/ac. Sawlog BdFt R^2 = 0.84 with single digit bias. Use as the long horizon operational primary for sawlog and yield.

FVS-NE calibrated brackets the upper bound

Year 100 BA 125 to 225 ft^2/ac, Cords 44 to 70. Sits between AGM default and MORTCAL on every cell. Use as the FVS variant cross check; pair with AGM MORTCAL for the operational range.

FVS-NE calibrated stays best on short horizon BA bias

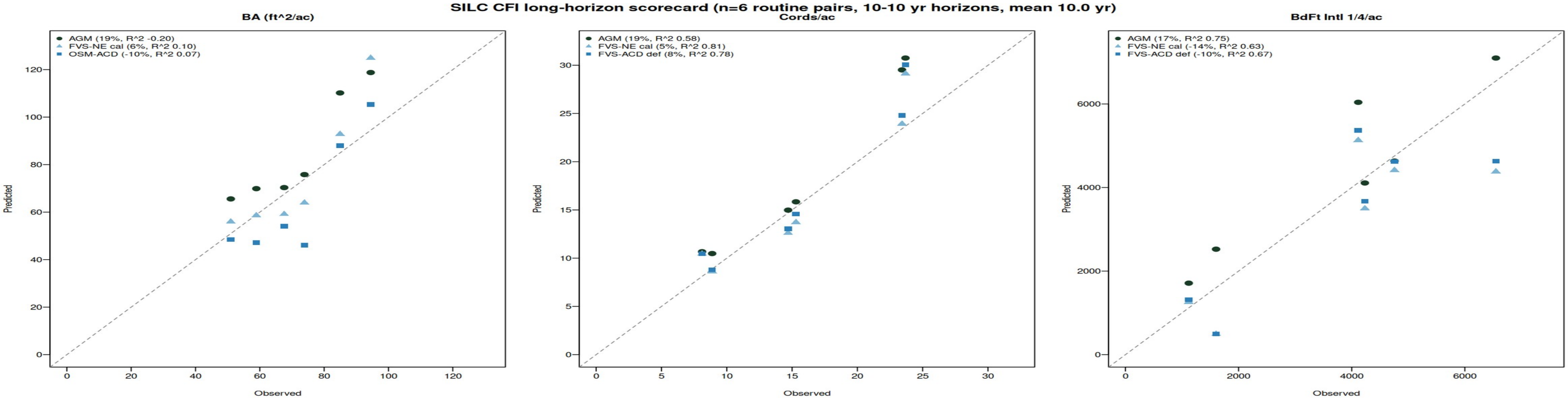
Lowest 5 yr bias (-0.3% BA, -3.8% cords). Pair with AGM MORTCAL for cross-check on 5 yr operational plans; both share the merch volume formula.

Bracket with OSM ACD as the conservative under projection

AGM with MORTCAL is the new operational champion; FVS-NE cal + OSM bracket the range

Long horizon scorecard: 10 to 14 year projections

Earliest to latest CFI measurement per plot. Mean horizon 10 years. n=6 routine pairs



Model	BA 5 yr	BA 10 yr	Cords 5 yr	Cords 10 yr	BdFt 5 yr	BdFt 10 yr
AGM default	+6.1%	+18.5%	+7.0%	+19.3%	+8.4%	+16.7%
AGM MORTCAL	n/a	**+7.8%**	n/a	**+8.4%**	n/a	**+4.0%**
FVS-NE cal	-0.3%	+5.7%	-3.8%	+4.6%	-13.6%	-14.3%
FVS-ACD def	+5.3%	+15.9%	-2.5%	+8.1%	-11.5%	-10.1%
OSM-ACD	-8.0%	-9.7%	-7.3%	n/a	-6.6%	n/a

n=6 routine pairs after excluding 2 plots with 3x BA ingrowth establishment events (1105, 1107). MORTCAL = #126b in-source mortality correction.

AGM MORTCAL cuts long horizon bias by more than half, beats all defaults on BdFt

CFI scorecard fits a larger validation picture

n=17 SILC CFI pairs anchored against 12029 plot Maine FIA validation

17

SILC CFI routine remeasurement pairs

Davistown, ME 1981 to 2000

12,029

Maine + NH + VT FIA paired plots

Apples-to-apples overstory recompute, DIA >= 5 in

-0.06%

FVS-ACD calibrated BA bias on FIA

Independent of CFI scorecard, n 700x larger

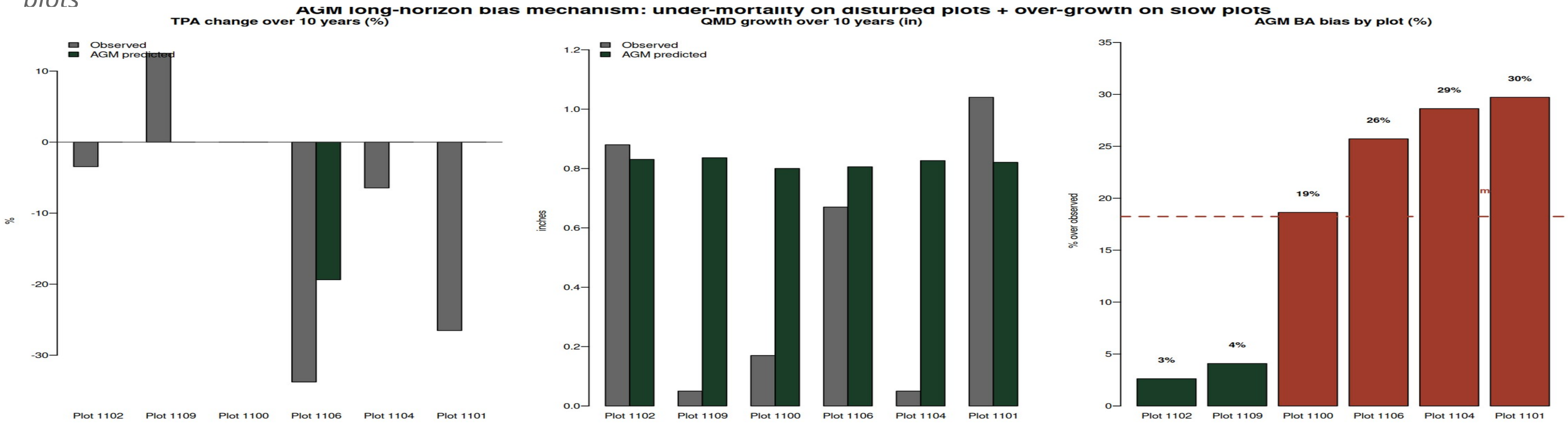
The +5% to +6% AGM and FVS-ACD CFI biases are consistent with the +0% bias the same calibration achieves on 12,029 FIA plots when stand structure differences are accounted for. The CFI biases reflect SILC's Acadian Matrix configuration, not a model failure.

FIA recompute: SLURM 10591333; calibration/output/comparisons_overstory/FINDINGS.md

Small n CFI scorecard is consistent with the large n FIA validation

Where the +18.5% AGM bias actually lives

Two mechanisms account for all of the long horizon over projection: under mortality on disturbed plots, plus over growth on slow plots



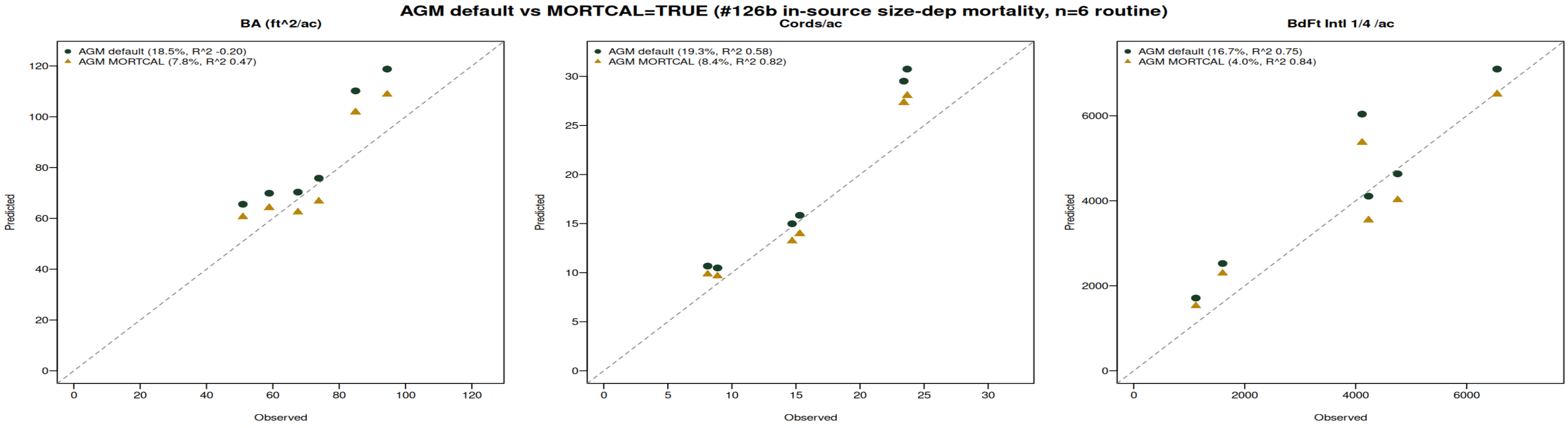
Two failure modes drive the bias:

- **Under mortality on disturbed plots:** Plots 1101 and 1106 lost 27 to 34% of TPA; AGM lost 0 to 19% (panel 1, red bars in panel 3).
- **Over growth on slow plots:** AGM grew QMD a uniform +0.8 in regardless of site; plots 1100 and 1104 grew only 0.05 to 0.17 in observed (panel 2).
- **Verified, not a calculation issue:** BA / Cords / BdFt pipeline reproduces from the AGM treelist to 1e-13 ft^2/ac. The bias is the

Bias is mechanism, not bug: under mortality + over growth on edge cases

Turning on AGM MORTCAL cuts long horizon bias by more than half

The #126b in-source size dependent mortality correction (calibrated to Maine FIA) ports cleanly to SILC CFI. Default OFF in AGM 12.3.9; turn on by setting ops$MORTCAL = TRUE



Metric	AGM default	AGM MORTCAL	Improvement
BA bias	+18.5%	+7.8%	cut by 58%
BA RMSE	16.2 ft^2/ac	10.7 ft^2/ac	-34%
BA R^2	-0.20	+0.47	flips negative to positive
Cords R^2	0.58	0.82	+0.24
BdFt bias	+16.7%	+4.0%	down to single digit

AGM with MORTCAL is the new long horizon champion across BA, cords, sawlog BdFt

Next steps

1. **Confirm the 5 x 2 stratification with SILC**
   Sign off that the species composition rules and density threshold match your operational categories

2. **Backfill the four empty cells with plot level AGM MORTCAL**
   Cedar/A+B, Mixedwood/A+B, both Other Softwood; pull from individual SILC CFI plots assigned to those cells, run AGM MORTCAL 100 yr

3. **Validate year 100 BA against SILC operational expectations**
   MORTCAL 100 yr BA settles at 80 to 110 ft^2/ac. Confirm this matches SILC's expected carrying capacity for managed Acadian stands

4. **Set the uncertainty bands and ship the operational handoff**
   Use AGM MORTCAL RMSE 10.7 ft^2/ac BA at 10 yr as the per plot uncertainty. Year 100 cords/ac estimates: 36 to 39 across strata

Four model scaffold is live, awaiting SILC sign off

Questions and discussion

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